

Pratham Arora

✉ pratham3992@gmail.com [in LinkedIn](#) [GitHub](#) [Portfolio](#)

Education

Plaksha University <i>B.Tech. in Computer Science & AI</i>	<i>Mohali, India — Expected Aug. 2026</i> <i>CGPA: 8.19 / 10.0</i>
Delhi Public School Ghaziabad Vasundhara <i>CBSE Class XII (PCM with Computer Science)</i>	<i>Ghaziabad, India — July 2022</i> <i>Percentage: 89.2%</i>
Delhi Public School Ghaziabad Vasundhara <i>CBSE Class X</i>	<i>Ghaziabad, India — July 2020</i> <i>Percentage: 98.0%</i>

Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, C#, HTML/CSS, MATLAB, Bash
AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenCV, SAM, GANs
Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST, GraphQL, Tailwind CSS
Cloud & MLOps: Azure (Cosmos DB, OpenAI), Docker, Git, Linux, Firebase, RAG Pipelines
Hardware: Arduino, Raspberry Pi, ESP32, Embedded Systems Integration

Research Experience

Undergraduate Researcher <i>Supervised by Prof. Sandeep Manjanna Plaksha University</i>	<i>June 2023 – Aug. 2023</i>
- Developed computer vision pipelines for semantic segmentation of agricultural imagery to differentiate crops from weeds. - Fine-tuned the Segment Anything Model (SAM) for the novel domain of crop imaging, improving segmentation Mean IoU (mIoU) by ~12% on sparse datasets. - Optimized image processing algorithms in Python, reducing data pipeline runtime by 40% for large-scale datasets.	

Industry Experience

AI Intern <i>Cotality Kolkata</i>	<i>June 2025 – July 2025</i>
- Architected a scalable FastAPI backend to automate code documentation, handling Abstract Syntax Tree (AST) analysis across 15+ languages. - Designed optimized NoSQL document schemas in Azure Cosmos DB, selecting appropriate partition keys to balance read/write throughput and reduce Request Unit (RU) consumption. - Designed RESTful API endpoints for the generative AI system, ensuring low-latency data retrieval for the frontend.	
Software Development Engineer Intern <i>Orangewood Labs Noida</i>	<i>June 2024 – July 2024</i>
- Refactored the monolithic RoboGPT codebase into a modular architecture, reducing code dependency by 40% and improving maintainability for complex food preparation tasks. - Integrated a Computer Vision stack with the control loop, optimizing calibration routines to significantly reduce robotic arm pose detection errors.	
Freelance Web Developer <i>Client Project Remote</i>	<i>May 2025 – June 2025</i>
- Engineered a digital footprint on the Vercel free tier, implementing SEO best practices and reducing hosting costs to zero. - Architected a modular Firebase CMS, empowering non-technical stakeholders to manage dynamic content without code intervention.	
Freelance Tutor <i>Python Programming Remote</i>	<i>June 2023 – May 2024</i>
Delivered personalized Python instruction to students, designing interactive lesson plans that enhanced engagement and comprehension of fundamental concepts.	

Selected Technical Projects

Reddit Data Ingestion Pipeline | *Pre-eclampsia Research* | Python, PRAW, VADER, LDA, Firebase Aug. 2025 – Dec. 2025

- Engineered a robust data ingestion pipeline using **PRAW** to scrape 22,000+ interactions from 19 Reddit communities.
- Designed a **NoSQL** schema in Firebase with real-time deduplication logic to maintain dataset integrity across concurrent scrapes.
- Identified a "Buffer Effect" (19.4% sentiment shift) and quantified a 35.8x post-pandemic volume increase in health communities.

VR LLM Conversational Agent | Unity, C#, Gemini 2.5 Flash, Google Cloud STT/TTS Aug. 2025 – Dec. 2025

- Built an immersive VR-based embodied conversational agent integrating large language models with real-time speech interfaces.
- Designed a controlled experimental framework to study the effect of latency feedback (gestures, fillers, visual indicators) on user experience.
- Conducted a within-subjects evaluation ($N = 18$) and analyzed animacy, anthropomorphism, and perceived responsiveness.
- Demonstrated that embodied gestures significantly improve immersion compared to explicit system indicators.
- Produced a publication-ready research paper on component-level latency feedback mechanisms in social VR.

Resume Builder & Interview GPT | Next.js, Tailwind, Firebase, Gemini API Jan. 2025 – April 2025

- Developed a production-ready full-stack application featuring an AI-powered resume generator with real-time PDF preview.
- Implemented persistent user sessions using **Client-Side Storage (LocalStorage)** and secured authentication flows to maintain state across browser reloads.
- Engineered an "Interview GPT" module to provide users with DSA guidance and personalized interview skill coaching.

UI Wireframe Generator | Python, PyTorch, Meta LLaMA, Computer Vision August 2024 – Dec. 2024

- Pioneered a novel approach by modifying a Self-Attention GAN (SAGAN) into a Cross-Attention GAN (CAGAN) to generate UI wireframes conditioned on textual prompts.
- Fine-tuned a LLaMA model on a custom dataset of segmented images and labeled JSON to translate natural language into structured UI data.
- Conducted a comparative analysis of generative models (DDPM, Stable Diffusion), evaluating their suitability for this specialized task.

Kelp Segmentation | Python, PyTorch, U-Net, Satellite Imagery Jan. 2024 – May 2024

- Designed a **U-Net** architecture for semantic segmentation of kelp forests in noisy satellite imagery, achieving a performance rank in the **Top 100 out of 600+** competitors.
- Engineered a pre-processing pipeline using custom CNN filters to remove atmospheric noise and improve label quality.

C-VRP with Time Window Constraints | Python, Google OR Tools, Google Maps API Sept. 2023 – Dec. 2023

- Implemented an optimization solution for the Capacitated Vehicle Routing Problem (VRP) with time window constraints using Google OR-Tools.
- Integrated real-world transit data via the Google Direction Matrix API to generate practical, efficient, and accurate routing solutions.
- Developed route visualizations using the Google Maps API for intuitive interpretation of results.

C Compiler & Coding Teaching Platform | Next.js, TypeScript, AST Visualization Jan. 2025 – May 2025

- Designed an interactive, Scratch-inspired platform to teach C programming concepts visually.
- Implemented real-time Abstract Syntax Tree (AST) visualization to demystify how source code is parsed and processed by a compiler.

Basketball Scoreboard & Data Tracker | Next.js, TypeScript, Firebase Dec. 2023 – Feb. 2025

- Architected a real-time basketball scoreboard with a focus on sub-100ms latency for live-game scorekeeping.
- Iteratively refined the UI based on 6 months of stress testing, incorporating user feedback to enhance operational speed and accuracy.

Sports Fest Site | Next.js, TypeScript, Framer Motion Jan. 2025 – Feb. 2025

- Engineered a fully responsive and performant website for Eklavya '25, the university's annual sports fest, utilizing Framer Motion for fluid animations.

Linux How-To & Bash CheatSheet | HTML/CSS, JavaScript, GitHub Pages Dec. 2023 – Feb. 2024

- Authored comprehensive technical documentation for essential Linux and Bash commands, designing a user-friendly interface with search and visualizations.

Weighbridge | *Arduino Uno, Hardware Integration, Embedded Systems*

Aug. 2023 – Dec. 2023

- Co-engineered an Arduino-based weighbridge system, integrating load cells and an LCD display to accurately measure vehicle weight.

Leadership & Involvement

Head of Tech

April 2025 – Present

Athleda (Plaksha Sports Society) | Plaksha University

- Overseeing the end-to-end development lifecycle of the main society website, managing project architecture and team workflow.

Head of Web Development

Feb. 2024 – Feb. 2025

LEAP.AI@Plaksha | Plaksha University

- Led the web development team, spearheading the design and maintenance of the AI club's official website.
- Mentored junior developers in web development best practices using Next.js and Vercel, fostering team growth.

Tech Team Member

April 2024 – Sept. 2024

E-Cell@Plaksha | Plaksha University

- Developed and maintained the entrepreneurship cell's website and multiple event-specific pages, including a club merchandise store.

Head of Procurement

Jan. 2024 – Dec. 2024

Plaksha Baking Club | Plaksha University

- Managed the end-to-end procurement process, executing market research and vendor negotiations to optimize the budget.